MODULE 11 · DOMAIN 1 - DATA PREPARATION FOR ML AND AI

Data for fine-tuning and distillation

Three techniques, three completely different training files.

MODULE 11 of 43 DOMAIN 1 - Data Prep WEIGHT 28%

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The **shape of the training file** tells you which technique you are doing.

What each one changes, and what it needs

TECHNIQUE	IT TEACHES, AND IT NEEDS
Fine-tuning (SFT)	Behaviour and format. Needs labelled prompt-completion pairs
Continued pre-training	Domain language and knowledge. Needs a large unlabelled corpus
Distillation	A teacher's behaviour, cheaply. Needs prompts only

ON THE EXAM

Retrieval is the fourth option, and usually the first one to consider. Module 01's rule: knowledge problem, retrieve.

The file format for supervised fine-tuning

- **JSONL.** One JSON object per line, one training example per line.
- Each line has a **prompt** field and a **completion** field. Both are required.
- The completion is the **ideal output**: the exact answer, in the exact format you want.
- Stored in Amazon S3, and referenced when you create the customisation job.
- Newer models use the **Converse API format** instead, with structured conversation turns.

> ON THE EXAM
>
> Prompt-completion is the classic format. Check the model's requirements, because several now expect Converse format.

Validating prompt-response pairs

- **Every completion must be a good answer.** The model learns from all of them, including the bad ones.
- One badly formatted completion in fifty teaches the model that format is optional.
- **Consistency of format** matters more than variety. Same structure in every completion.
- Check that each completion actually answers its own prompt. Mismatched pairs are common in scraped data.
- Deduplicate. Repeated pairs make the model overconfident about that answer specifically.

> ON THE EXAM
>
> 500 excellent examples beat 5,000 noisy ones. This is the one place where curation beats collection outright.

Content safety screening

- **The model will reproduce what you teach it.** Including tone you did not intend.
- Screen for **personal data**, using Macie from module 08. Fine-tuned weights can leak training examples.
- A model cannot be un-trained. Removing a record after the fact does not remove its influence.
- Screen for toxicity, bias and anything legally sensitive, before the job runs.
- Screen for **outdated facts** too. Fine-tuning bakes them in, where retrieval would not.

> ON THE EXAM
>
> Fine-tuning data is the one dataset where a single bad record can survive into every future response.

Volumes, and what actually helps

- A few hundred good examples can already improve a model measurably.
- Titan fine-tuning accepts **up to 10,000 training records**. Quotas vary by model.
- The relevant quota is the **sum of training and validation records**, not training alone.
- More data does not fix inconsistent data. It teaches the inconsistency more thoroughly.
- **Cover the edge cases.** Tricky inputs are what fine-tuning is bought for.

> ON THE EXAM
>
> If the model already handles the common cases, adding a thousand more common cases changes very little.

Token limits, and what they force

- Each record has a maximum token count, and it varies by model. Check before exporting.
- An over-long record is **rejected or truncated**, and truncation cuts the end of the completion.
- So the part that gets lost is usually the sign-off, the conclusion, or the final step.
- Long documents belong in retrieval, not in a fine-tuning example.
- Count tokens before submitting. A character count is a poor proxy for a token count.

> ON THE EXAM
>
> Truncation is worse than rejection, because a rejected job tells you and a truncated record does not.

Where fine-tuning data actually comes from

SOURCE	WHAT YOU GET
Historical support tickets	Real prompts with real expert answers. Usually the best source
Invocation logs	Real prompts. Completions only if a model already answered well
Existing documentation	Convert procedures into question and answer pairs
Expert authoring	Highest quality, slowest, and the most expensive
Synthetic generation	A larger model writes examples. This is distillation by another name

ON THE EXAM

Most organisations already hold better fine-tuning data than they realise, sitting in ticket systems and inboxes.

The validation dataset

- A separate JSONL file in the same format, held out from training.
- It gives you **validation loss per epoch**, which is how you detect overfitting.
- Training loss falling while validation loss rises means **stop**. Take the earlier checkpoint.
- Roughly a 10 to 20 percent holdout is typical. It counts against your record quota.
- Distillation is different: Bedrock splits the augmented dataset for you.

> ON THE EXAM
>
> Module 05's rule holds here too. Held-out data must be genuinely held out, or the loss curve tells you nothing.

Fine-tuning is iterative

- First run establishes a baseline. It is rarely the version you ship.
- Evaluate, find the failure cases, and **add examples that target them specifically**.
- Targeted additions beat bulk additions. Twenty examples of a failure mode change more than two thousand random ones.
- Each iteration costs a full training run, so batch your fixes rather than retraining per idea.
- Keep the evaluation set fixed across runs, or you cannot compare them.

> ON THE EXAM
>
> Same discipline as module 05: change one thing, measure against a fixed benchmark, and keep the comparison honest.

Reproducibility of a customisation run

- Version the **exact training file**, not the pipeline that produced it.
- Record the base model and version, the hyperparameters, and the job identifier.
- 'Which data produced this model?' is asked by auditors and by incident reviews alike.
- A model whose training data you cannot reproduce cannot be re-created or defended.
- Module 06's dataset versioning and module 27's model cards are where this is recorded.

> ON THE EXAM
>
> For a fine-tuned model the training file IS the explanation of its behaviour. Losing it loses the explanation.

A corpus, not examples

- JSONL again, but each line has **only a prompt field**. No completion.
- That is the whole difference in the file, and it is the fastest way to tell them apart.
- It teaches **vocabulary, phrasing and domain knowledge**, not how to answer a question.
- Accepts far more data: up to **100,000 records** for Titan, and it wants volume.
- AWS notes positive effects typically appear after around a billion tokens of domain text.

> ON THE EXAM
>
> Continued pre-training makes a model fluent in your domain. It does not make it obedient. That still needs fine-tuning.

Fine-tuning or continued pre-training?

FINE-TUNE, WHEN

You have examples

- You know the ideal output for each input.
- You need a specific format or tone.
- Hundreds to thousands of pairs.
- The task is narrow and well defined.

CONTINUED PRE-TRAIN, WHEN

You have a corpus

- You have text, and no ideal answers.
- The domain vocabulary is unfamiliar to the model.
- Tens of thousands of documents.
- You want broad competence, not one task.

THE TIEBREAKER

Do you have answers, or just text? That question separates them, and the file format follows immediately.

How a teacher trains a student

- You choose a **teacher** whose quality you want, and a **student** that is smaller and cheaper.
- You supply **prompts only**. The teacher generates the responses.
- Those prompt-response pairs then fine-tune the student. It is SFT with generated labels.
- Bedrock **augments the dataset** with data-synthesis techniques, then splits train and validation itself.
- An AWS example pairs Nova Premier as teacher with Nova Micro as student.

> ON THE EXAM
>
> The reported gain on that example: inference cost cut by over 95 percent and latency halved, on one specific task.

Preference data: RLHF and DPO

- Humans **rank or compare** two model responses rather than writing an ideal one.
- Far cheaper to produce than ideal answers. Judging is easier than authoring.
- **RLHF** trains a reward model from those preferences, then tunes against it.
- **DPO**, direct preference optimisation, skips the reward model and tunes on the pairs directly.
- This is the labelling work module 07 described as the FM-era shift.

ON THE EXAM

Preference data suits judgements nobody can write down: helpfulness, tone, or which of two answers a customer would prefer.

Invocation logs as distillation data

- Bedrock can use your **invocation logs** as the prompt source, not just a curated file.
- Those are **real production prompts**, in the real distribution your users generate.
- No labelling effort at all, because the teacher supplies the answers.
- Logs may contain personal data. The screening slide applies here too.
- It also means you must have enabled invocation logging before you needed it.

> ON THE EXAM
>
> This is the cheapest high-quality training data in the whole course: prompts you already received, answered by a model you already trust.

Distillation or ordinary fine-tuning?

DISTIL, WHEN

The big model already works

- A large model gives the quality you want.
- It is too slow or too expensive to keep using.
- You have prompts, but no ideal answers.
- The task is narrow enough to transfer.

FINE-TUNE, WHEN

No model does it yet

- No off-the-shelf model produces what you need.
- You have human-written ideal answers.
- The output style is yours, not a model's.
- Domain judgement no teacher model has.

THE TIEBREAKER

Distillation copies a model. Fine-tuning copies humans. If nothing currently does the task well, there is no teacher to copy.

Catastrophic forgetting

- Training hard on a narrow task makes a model **worse at everything else**.
- A model fine-tuned only on ticket classification may lose general conversational ability.
- **Mitigate with data:** mix in general examples, do not train only on the narrow task.
- Fewer epochs and a lower learning rate reduce it. That is module 15's territory.
- Evaluate on general tasks too, not only on the task you tuned for.

> ON THE EXAM
>
> The metric you optimised will look excellent. The capability you lost is only visible if you thought to measure it.

Serving a customised model

- **Provisioned Throughput** gives dedicated capacity. One model unit needs no commitment.
- Beyond one unit requires a **one-month or six-month commitment**. That is a real decision.
- **On-demand inference on custom models** now exists, but not for everything.
- For Nova, it covers custom **understanding** models trained **after 16 July 2025**. Not content generation.
- For SageMaker AI customisation, on-demand covers **PEFT** models only, not full-rank fine-tuning.

> ON THE EXAM
>
> Plenty of material still says custom models always require Provisioned Throughput. That is no longer universally true.

Full fine-tuning and PEFT

- **Full-rank fine-tuning** updates every weight. Most powerful, most expensive, most forgetting.
- **PEFT** trains a small set of added parameters and freezes the rest. **LoRA** is the common form.
- PEFT is cheaper, faster, and **far less prone to catastrophic forgetting**.
- It also decides your serving options: on-demand inference on SageMaker-customised models is **PEFT only**.
- Several adapters can share one base model, which is how multi-tenant customisation works.

> ON THE EXAM
>
> A training-time choice that constrains a deployment-time option. Those are worth noticing, because questions build on them.

What customisation costs

COMPONENT	COST SHAPE
Fine-tuning job	Customisation pricing, by tokens processed and epochs
Continued pre-training	The same, over a much larger corpus. So much larger
Distillation, step 1	Teacher generation charged at the teacher's on-demand rate
Distillation, step 2	Student training at customisation pricing
Serving, Provisioned	Per model unit-hour. First unit uncommitted, then a term commitment
Serving, on-demand	Per token, where the model and training date support it
Doing it again	Every re-run repeats the whole training cost

ON THE EXAM

Distillation costs twice at training time and pays back at inference time. The payback needs volume to arrive.

Security around a customisation job

- Training data sits in **S3**, encrypted with SSE-S3 or a customer-managed KMS key.
- The **custom model artefact** can be encrypted with your own KMS key too.
- Customisation jobs can run with a **VPC configuration**, keeping traffic off the public internet.
- An IAM service role grants the job read access to data and write access to output.
- AWS does not use your customisation data to train base models for anyone else.

> ON THE EXAM
>
> The output artefact is as sensitive as the input data, because it now contains what the data taught it.

Symptoms, and what they usually mean

SYMPTOM	USUAL CAUSE
Job rejected before training starts	Wrong file shape for the technique, or malformed JSONL
Model output format is inconsistent	Completions in the training file were not consistently formatted
Model is great at the task, poor at everything else	Catastrophic forgetting. Mix in general examples
Validation loss rising while training loss falls	Overfitting. Stop early, take the better checkpoint
Answers are confidently out of date	Facts baked in by fine-tuning. Should have been retrieval
Record count rejected as over quota	Quota is training plus validation records combined
Custom model cannot use on-demand inference	Model type or training date outside the supported set

ON THE EXAM

The fifth row is the expensive mistake, because it only becomes visible once the facts have changed.

Fine-tuning teaches **how to behave**. It is an expensive way to teach **what is true**.

A support team has 800 past tickets, each paired with the reply a senior agent actually sent. They want a model to draft replies in the same structure: greeting, diagnosis, numbered steps, sign-off.

The information needed to answer is already available to the model through an existing knowledge base. Only the format and tone are wrong.

What should they do?

A Continued pre-training on the 800 tickets as an unlabelled corpus

B Fine-tune with a JSONL file of prompt and completion pairs from the 800 tickets

C Distillation, using a larger model as teacher over the 800 tickets

D Add the 800 tickets to the knowledge base so the model can retrieve them

◆ PAUSE AND DECIDE

What should they do?

A. Continued pre-training on the 800 tickets as an unlabelled corpus

Throws away the labels. It teaches vocabulary rather than behaviour, and 800 records is far below the volume CPT wants.

B. Fine-tune with a JSONL file of prompt and completion pairs from the 800 tickets

A behaviour problem with ideal answers already written. 800 pairs is a workable volume, and the senior replies are the completions.

C. Distillation, using a larger model as teacher over the 800 tickets

Would copy a teacher model's style, and the wanted style is the senior agents'. The ideal answers are human-written, so there is nothing to distil.

D. Add the 800 tickets to the knowledge base so the model can retrieve them

Retrieval supplies facts, and facts are stated as already available. Format and tone are not retrieved.

THE SIGNAL

Two facts settle it: the information is already available, and ideal answers already exist. Behaviour problem, with labels in hand.

A marine insurance firm finds that models handle its documents poorly. Terms like 'general average', 'protection and indemnity' and 'demurrage' are misread as everyday English, producing confident nonsense.

They hold roughly 60,000 policy documents, surveyor reports and claim files. Nobody has written ideal answers for any of them, and there is no budget to do so.

Which technique fits?

A Fine-tuning, using document titles as prompts and bodies as completions

B Distillation from a larger model over the 60,000 documents

C Retrieval over the 60,000 documents with a knowledge base

D Continued pre-training on the 60,000 documents as an unlabelled corpus

◆ PAUSE AND DECIDE

Which technique fits?

A Fine-tuning, using document titles as prompts and bodies as completions

Manufactures pairs that were never a question and an answer. It teaches the model to generate documents from titles, which nobody asked for.

B Distillation from a larger model over the 60,000 documents

Needs a teacher that already handles the domain. The premise is that available models misread these terms, so no teacher exists.

C Retrieval over the 60,000 documents with a knowledge base

A reasonable step, and it supplies passages without fixing comprehension. A model that misreads 'general average' misreads it in a retrieved chunk too.

D Continued pre-training on the 60,000 documents as an unlabelled corpus

A vocabulary problem with a large unlabelled corpus and no ideal answers. That is precisely what CPT is for, and 60,000 is the right scale.

THE SIGNAL

This is the rare case where retrieval is not the answer. The model does not lack the document; it misunderstands the language in it.

A company routes incoming requests using a large, expensive model. Quality is excellent and everyone is satisfied with the routing decisions. Inference cost and latency are now the problem at production volume.

They have twelve months of invocation logs containing real user prompts. Nobody has labelled the correct routing decision for any of them.

What should they do?

A Continued pre-training a small model on the invocation logs

B Fine-tune a small model using the logs as prompts and completions

C Distil the large model into a smaller student, using the logs as prompts

D Add prompt caching and accept the current model and cost

◆ PAUSE AND DECIDE

What should they do?

A Continued pre-training a small model on the invocation logs

Teaches vocabulary, and vocabulary is not the problem. Quality is already excellent; only cost and latency are wrong.

B Fine-tune a small model using the logs as prompts and completions

The logs hold prompts. There are no labelled completions, so there is nothing to put in the completion field without labelling twelve months of data.

C Distil the large model into a smaller student, using the logs as prompts

A model whose quality you want, prompts but no labels, and cost as the problem. Bedrock can take invocation logs directly as the prompt source.

D Add prompt caching and accept the current model and cost

Caching helps with repeated prefixes and does not change the per-request model cost for varied user prompts. It is a partial measure at best.

THE SIGNAL

Every precondition for distillation is present: an existing model that works, prompts without labels, and cost as the stated problem.

> A team is ready to fine-tune on 4,000 prompt-completion pairs exported from a shared inbox. Spot checks find customer names and account numbers in some completions, several replies that answer a different question from their prompt, and three different sign-off formats.
>
> The delivery date is in four days and the team proposes running the job now and cleaning the data afterwards.

How should you respond?

A. Screen and validate the pairs before training, because a trained model cannot be un-trained

B. Run the job now, then run a second fine-tuning job on cleaned data to correct it

C. Run the job now and add a Guardrail to filter personal data from responses

D. Run the job now, since 4,000 examples will average out the inconsistencies

◆ PAUSE AND DECIDE

How should you respond?

A Screen and validate the pairs before training, because a trained model cannot be un-trained

All three defects are training-data defects. Screening for personal data, checking pairs match, and standardising format all have to happen first.

B Run the job now, then run a second fine-tuning job on cleaned data to correct it

Training again does not remove what was learned, and it costs the full job twice. The personal data remains in the weights either way.

C Run the job now and add a Guardrail to filter personal data from responses

A useful control that sits at inference. It does not stop the data entering the weights, and it does nothing for mismatched pairs or format.

D Run the job now, since 4,000 examples will average out the inconsistencies

Backwards. Three sign-off formats teach the model that all three are acceptable, so inconsistency is exactly what it learns.

THE SIGNAL

Screening is not a step you can defer, because there is no undo. That is what makes fine-tuning data different from every other dataset.

What to take from this module

- **Prompt and completion** means fine-tuning. **Prompt only** means continued pre-training, or distillation.
- **Do you have answers, or just text?** That question chooses between SFT and CPT.
- **Distillation copies a model. Fine-tuning copies humans.** No teacher means no distillation.
- **Screen before you train.** There is no un-training, so screening cannot be deferred.
- **Fine-tuning teaches behaviour.** Teaching facts this way means retraining when they change.

> ON THE EXAM
>
> That closes Domain 1, at 28 percent of the exam. Module 12 opens Domain 2: choosing the modelling approach.